

Giovanni Lunetta

Experienced Engineer, Data Engineering

 glunetta02@gmail.com  Pequannock, NJ  Github  Medium  LinkedIn  My Website

PROFESSIONAL SUMMARY

AI and data professional with 2+ years building production GenAI and ML systems in enterprise healthcare — now driving application strategy and AI-led process modernization within J&J's Acquisitions & Divestitures organization. Designed and shipped a fully autonomous agentic document platform with semantic mapping, confidence-aware automation, LLM governance, and 409 tests in under 4 weeks. Built multi-brand ML pipelines across 3 therapeutic areas and architected event-driven AWS pipelines eliminating 200+ hours of manual processing annually. Skilled in Python, LangChain, agentic workflow orchestration, RAG architectures, XGBoost, clustering, and AWS.

WORK EXPERIENCE

Experienced Engineer — Data Engineering, 05/2026 – Present | New Brunswick, NJ
Johnson & Johnson (Acquisitions and Divestitures)

Technology LDP Analyst — Data & Decision Science, 06/2024 – 05/2026 | Raritan, NJ
Johnson & Johnson (MedTech R&D)

- **Agentic Document Intelligence Platform:**

- Designed and delivered a fully autonomous agentic onboarding platform enabling business users to independently discover automation opportunities and generate production-ready extraction rules — without data science involvement. Two operating modes (Mapping Mode for source-to-template variable mapping; Explore Mode for template-free discovery), semantic-first mapping, and confidence-aware automation (AUTO_MAP vs. NEEDS_REVIEW).
- Engineered pipeline resumability via checkpointing, full LLM governance with per-call audit trails, and a Flask web UI with API key auth, SSE streaming, and rate limiting. CI/CD via GitHub Actions with pytest (409 tests) and Playwright E2E. Delivered production-hardened in 3–4 weeks.

- **Vision-Based PDF Document Extraction:**

- Built vision-based PDF table extraction using GPT-4o multimodal, rendering 400+ page regulatory documents as 300 DPI images paired with extracted text to capture complex tables with merged cells and hierarchical structures. Engineered TOC-based document segmentation (62 sections per CER) to fit LLM context windows. Achieved 90–95% accuracy vs. 70–80% text-only, reducing manual processing from 4+ hours to <30 minutes per document.

- **Persistency Modeling — Multi-Brand ML Pipelines:**

- Built a persistency model for a treatment-resistant depression therapy from scratch — integrating 2 fulfillment channels (82K patients, 3.9M patient-weeks), engineering 40 leakage-safe features across 7 clinical categories, and training XGBoost with Optuna tuning (ROC-AUC=0.769, F1=0.687, 16.7x lift). Designed phase-aware gap thresholds from synthesis of 6+ clinical studies, capturing 37% more patients than the vendor pipeline.
- Validated a 14-node Kedro ML pipeline for a biologic drug model (1.3M+ records, ROC-AUC=0.717), identifying a critical sampling bug that caused loss of validation/test splits — preventing erroneous evaluations from reaching production and producing comprehensive documentation for cross-brand replication.

- **Commercial ML Analytics — Surgical Preference & HCP Segmentation:**

- Delivered surgeon preference classification for a robotic-assisted surgery platform combining rule-based classification (67% coverage, +23 pts over baseline) with Fuzzy C-Means clustering (silhouette=0.66) across 4 feature sets. Built a Tableau dashboard used in clinical decision support and product strategy.
- Segmented ~10K HCPs treating a rare hematologic condition by engineering 90+ features, training statistically validated K-Means clustering, and delivering 6 personas adopted by the commercial team for targeted outreach. Extended into treatment pattern analysis (awarded as follow-on) to identify innovative prescribers vs. those needing clinical education.

Technology Summer Intern — Data Science,

05/2023 – 08/2023 | Titusville, NJ

Johnson & Johnson (Janssen Pharmaceuticals)

- Developed a machine learning model using healthcare claims data to predict patient drug eligibility 6 months in advance, optimizing doctor outreach targeting and pharmaceutical processes.

EDUCATION

M.S. in Data Science, *University of Connecticut*

09/2023 – 05/2024 | Storrs, CT

B.A in Stats & Poli Sci, Math (Minor), *University of Connecticut*

09/2020 – 05/2023 | Storrs, CT

PROJECTS

DateBack, *(macOS & Windows Application)*

- Built production macOS desktop app (Electron + Python) processing Snapchat exports (50GB+) with EXIF date restoration, batched cloud streaming, and Apple code signing. Full professional distribution (notarization, auto-updates, DMG installer) with 100% offline-first architecture.

AI Portfolio Assistant, *(Self-Evaluating LLM System)*

- Architected production conversational AI using GPT-4o-mini and Gemini 2.0 Flash featuring automated quality control, function calling for lead capture, and self-improving feedback loop. Implemented evaluation retry loop with dual-model assessment (generation vs. evaluation) and JSONL-based analytics for continuous improvement. Deployed via Gradio.

Hack4Health — SSCP Clinical Device Recommendation Agent

- Architected a 7-agent pipeline transforming EU MDR regulatory documents into explainable clinical device recommendations — dual-audience output (HCP + patient), 10 guardrail rules, keyword + LLM safety gate, multi-turn Flask UI with simulated EHR integration, and competitive intelligence analytics.

SKILLS

- **Agentic AI & LLM Engineering:** OpenAI API, GPT-4o multimodal, LangChain, LangGraph, RAG architectures, hybrid retrieval (vector + BM25), Postgres + pgvector (HNSW), agentic workflow design (stage orchestration, checkpointing, resumability), semantic search, document embeddings, schema-validated extraction, LLM governance/audit trails, prompt engineering, token optimization, vision-based document extraction, Flask (SSE streaming, API auth)
- **Machine Learning & Modeling:** Python, R, XGBoost, neural networks, Kedro pipelines, semi-supervised learning (co-training, self-training), clustering (Fuzzy C-Means, K-Means), PCA, evaluation (CV, AUC/PR, F1, silhouette, SHAP), ANOVA, Kruskal-Wallis, feature engineering, leakage-safe design
- **Data Engineering & Cloud:** AWS (Lambda, S3, Athena, Redshift, IAM, EventBridge, CloudShell), event-driven pipelines, ETL automation, SQL (Postgres, Redshift), Databricks (fundamentals), GitHub Actions CI/CD, pytest, schema design, query optimization, concurrent execution patterns
- **Analytics, BI & Product:** Tableau (Desktop Specialist certified), Power BI, agile delivery, stakeholder coordination, technical documentation

CERTIFICATES

- **Certified:** AWS Certified Cloud Practitioner | Tableau Desktop Specialist
- **Professional Programs:** Harvard/edX — Data Science Professional Certificate; CS50x | Babson Exec Ed — J&J Design Thinking
- **Specialized Training:** AI Engineer Agentic Track: The Complete Agent & MCP Course | DeepLearning.AI  — ML Specialization, Prompt Engineering, LangChain, Diffusion Models | Databricks Fundamentals Accreditation